

STEAM GAME ANALYTICS DASHBOARD

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Link: <https://steameda.streamlit.app/>





PROJECT GOALS



Identify Key Drivers of Popularity

Discover the characteristics that are associated with the most popular Steam games.



Explore Relationships

Examine relationships between price, ratings, player counts, ownership, and game genres.



Enable Interactive Exploration

Build an interactive dashboard that allows users to filter, analyze, and compare games dynamically.



Inform Game & Market Insights

Provide actionable insights for gamers, developers, and analysts interested in the Steam marketplace.



DATASET



Source:

SteamSpy API



Scope:

Top ~1000 Steam Games



Total Features:

18+ Key Attributes



Key Data Includes:

- Estimated Owners
- Price & Initial Price
- Positive / Negative Ratings
- Rating Score
- Concurrent Players (CCU)
- Game Tags / Genres
- Developers & Publishers
- Release Date & Platforms

DASHBOARD OVERVIEW

An interactive dashboard to explore what makes Steam games successful.

1 FILTERS

Use interactive controls to filter games by tags, price, rating, and more.

Apply Filters

Selecting multiple tags only shows games containing every selected tag.

Select Game Tags

Action FPS Multiplayer

Select all
Shooter
Classic
Team-Based
First-Person
Competitive
Tactical

Steam Data Analysis

The goal for this analysis is to better understand what makes some Steam games so much more popular than others by looking at trends and patterns within the top ~1000 steam games today.

Games

227

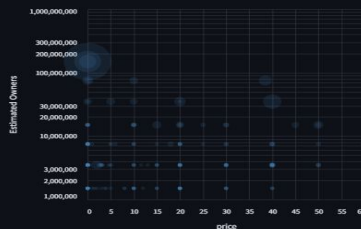
Median Owners

3500000

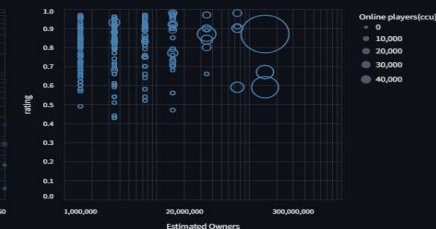
Median Price

\$9.99

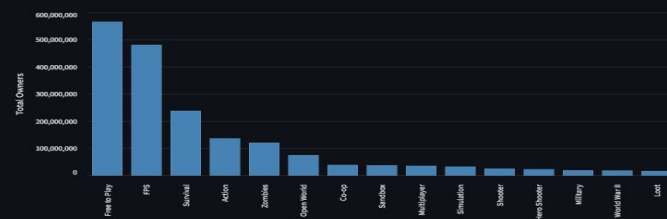
Game Owners Over Price



Game Rating Over Price



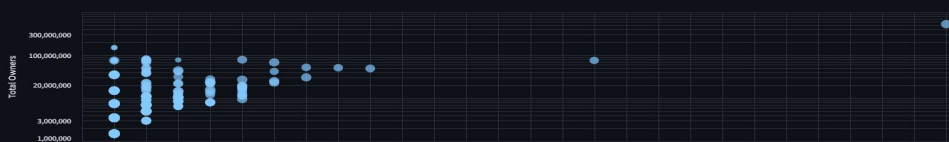
Top 15 Main Tags by Owners



Choose developer metric

Total Owners Median Owners

Total Owners vs Number of Games by Developer



2 LINKED VISUALIZATIONS

Selections in one chart automatically highlight related data in the other visualizations.

3 INTERACTIVE TAG CHART

Click a tag to highlight games with that primary tag across all visualizations.

4 DEVELOPER ANALYSIS

Compare developers based on the selected metric.



Side Bar and Filtering Tools



Apply Filters

Selecting multiple tags only shows games containing every selected tag.

Select Game Tags

Choose options



☐ Free games

Select desired price range



Select desired rating)



The dashboard includes several user interface controls that allow users to dynamically filter the dataset and focus on specific groups of games.



GAME TAG SELECTION

Select one or more tags to show games that contain all of the chosen tags.



PRICE RANGE SLIDER

Filter games within a specific price range.
Set to 0 to 59 dollars.



RATING RANGE SLIDER

Filter games by user rating (positive / all ratings).
Range from 0.00 to 1.00.



FREE GAMES FILTER

Toggle to show only free games (price = \$0).
Disables the price range slider when active.



DEVELOPER METRIC SELECTOR

Choose the metric used in the developer analysis chart
(Top by Median Owners, Median Rating, etc.).



WHY IT MATTERS

These controls help users quickly narrow down the data to answer specific questions and compare how trends change across different segments of Steam games.

Apply Filters

Selecting multiple tags only shows games containing every selected tag.

Select Game Tags

Choose options



☒ Free games

Select desired rating)

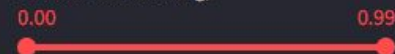


Chart Interactions

(Metrics responding to sidebar filters)

Interact directly with the charts to explore Steam games in real time. Your actions highlight and connect data across all visualizations on the dashboard.



BRUSH TO SELECT

Drag to select a region in either scatterplot. Selected games are highlighted across all other charts.



CLICK TO HIGHLIGHT

Click a bar in the tag chart to highlight games with that primary tag across the dashboard.



HOVER FOR DETAILS

Hover over any point to view detailed information about that game.



LINKED & SYNCHRONIZED

All interactions are linked—selections in one visualization update the others instantly.



WHY IT MATTERS

Linked visualizations reveal how ratings, pricing, popularity, and player engagement relate across Steam games.

Steam Data Analysis

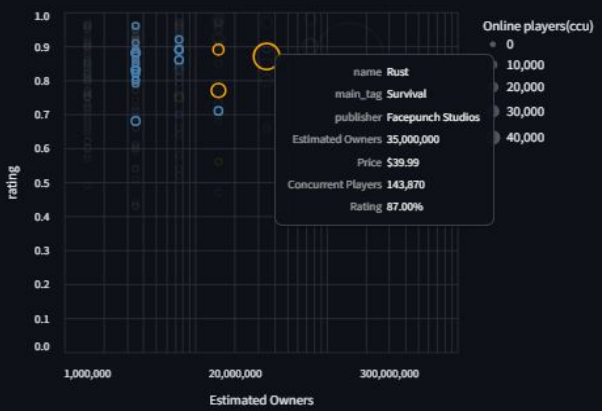
The goal for this analysis is to better understand what makes some Steam games so much more popular than others by looking at trends and patterns within the top ~1000 steam games today.

Games	Median Owners	Median Price	Median Rating
227	3500000	\$9.99	0.84

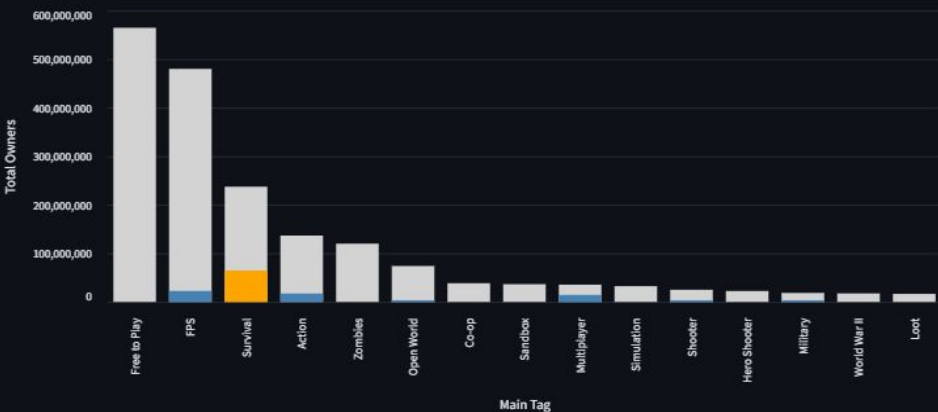
Game Owners Over Price



Game Rating Over Price



Top 15 Main Tags by Owners



KEY INSIGHTS

What the data reveals about successful Steam games.



RATINGS SHOW LIMITED CORRELATION WITH OWNERSHIP

Ratings show little relationship with ownership unless they fall below 50%, where games tend to have substantially fewer owners.



POPULAR GENRES DOMINATE OWNERSHIP

Free-to-play, Open World, and Survival games consistently rank among the most popular genres by both total and median ownership.



ENGAGEMENT FUELS LONG-TERM SUCCESS

Games with higher concurrent player counts (CCU) and positive feedback consistently perform better over time.



DESIGN TRADEOFFS

Every design choice comes with tradeoffs.



DESIGN DECISIONS



Logarithmic scaling was used for owner counts due to extreme variation between games.



Bubble size represents concurrent player counts.



Median values were used when appropriate to reduce the influence of outliers.



TRADEOFFS



SteamSpy provides **estimated ownership values** rather than exact sales figures.



Games may belong to **multiple tags** despite using a primary tag visualization.



Developer comparisons can be heavily influenced by **large publishers** with many releases.



This dataset includes only the **top 1,000 games on Steam**, so all insights reflect games that are already somewhat popular.



These tradeoffs should be considered when interpreting the results.

The insights are most valuable for understanding trends **among already successful Steam games**.



Baldur's Gate

STARDEW VALLEY

HADES